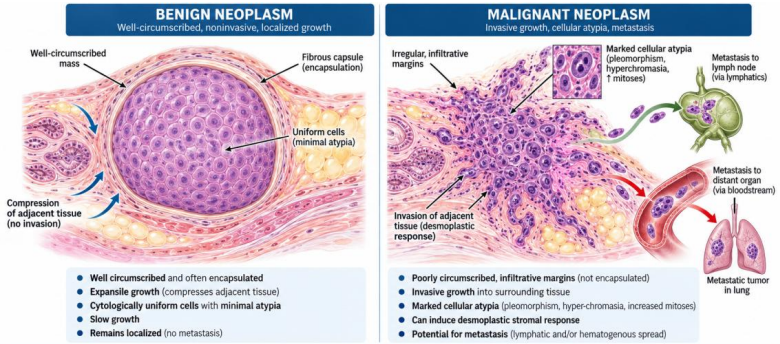
# Characteristics of tumors and epidemiology

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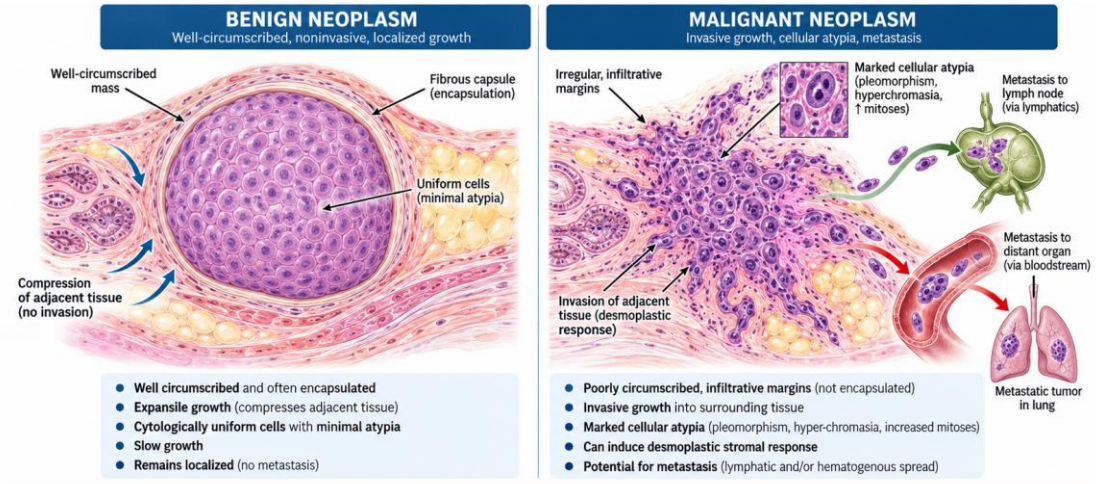
**Dr. Zheen Othman Hama Faraj**
**Lecturer of Histopathology**
**Department of Anatomy & Pathology, College of Medicine, HMU**




| | BENIGN NEOPLASM | MALIGNANT NEOPLASM |
|---|---|---|
| GROWTH RATE | Slow | Often rapid |
| MARGINS | Well circumscribed, often encapsulated | Poorly circumscribed, infiltrative |
| EFFECT ON ADJACENT TISSUE | Compresses (expansile) | Invades and destroys |
| CYTOLOGIC FEATURES | Uniform cells, minimal atypia | Marked atypia, pleomorphism, increased mitoses |
| METASTASIS | No | Yes (lymphatic and/or hematogenous) |

# Lecture outlines

- Characteristics of benign and malignant neoplasms
- Differentiation, anaplasia, dysplasia and carcinoma in situ
- Rate of growth, local invasion and metastasis
- Pathways of spread of malignant tumors
- Cancer epidemiology and major risk patterns




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## Characteristics of benign and malignant neoplasms:

### 1. Differentiation:

- The differentiation of parenchymal cells refers to the extent to which they resemble their corresponding normal cells morphologically and functionally.
- Benign neoplasms are composed of well-differentiated cells that closely resemble their normal counterparts.
- Malignant neoplasms are characterized by a wide range of parenchymal cell differentiation from:

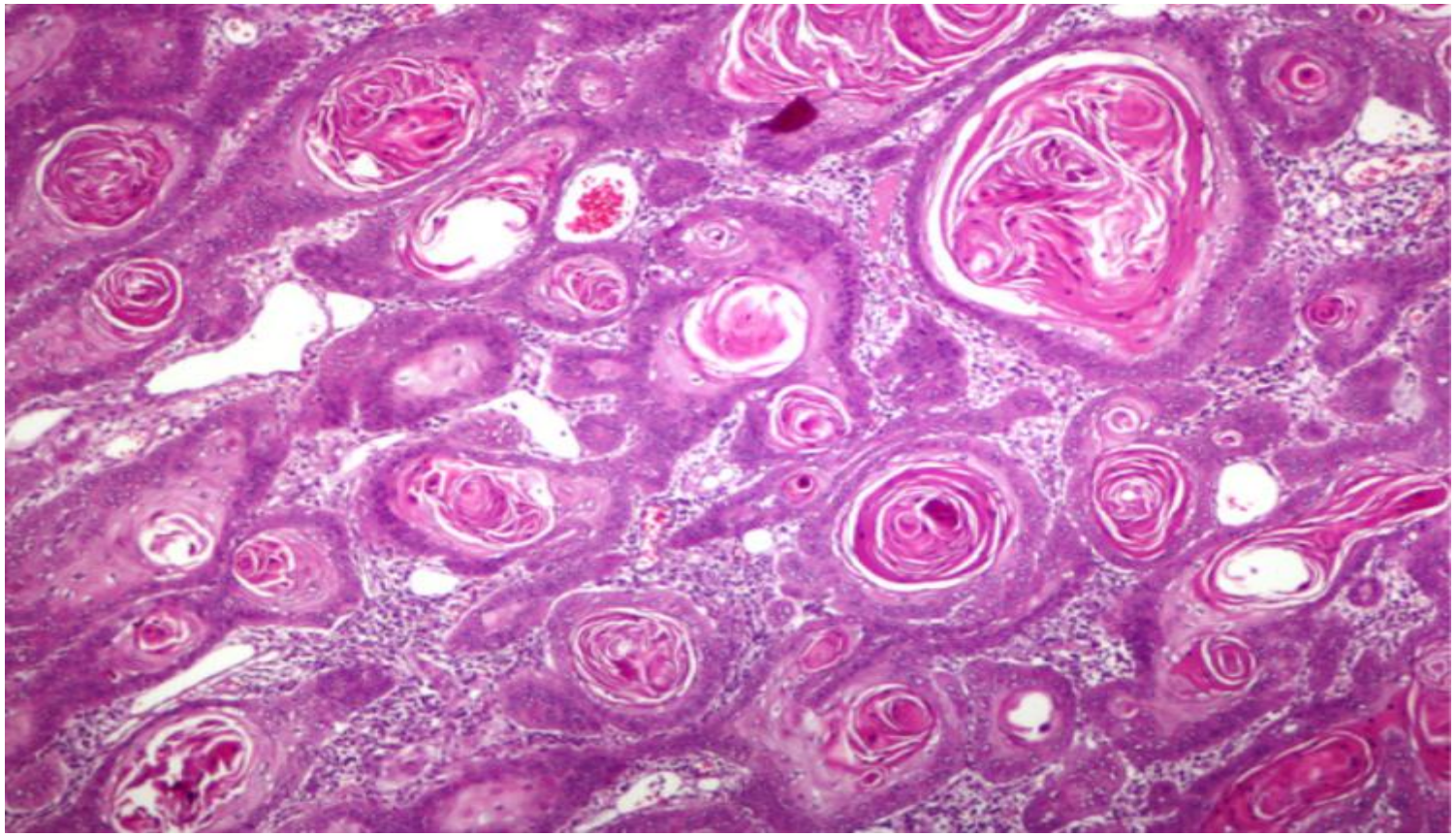
➢ Well differentiated
➢ Moderately differentiated
➢ Poorly differentiated (undifferentiated).

# Differentiation and anaplasia

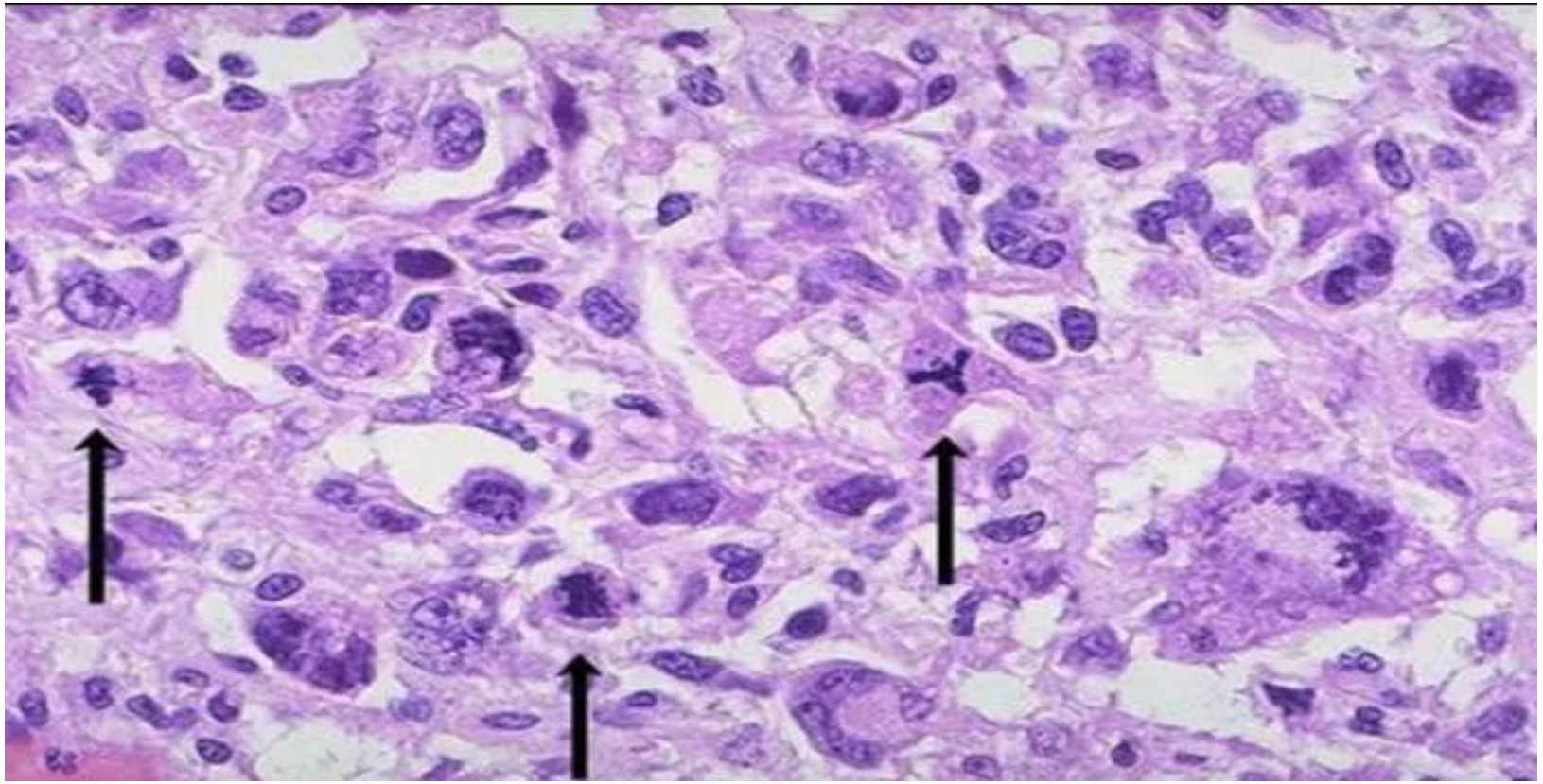
| Term | Meaning | Clinical/pathology importance |
|---|---|---|
| Differentiation | Extent to which tumor cells resemble normal cells morphologically and functionally | Basis for grading many tumors |
| Well differentiated | Cells closely resemble normal counterparts | Usually slower growth and better prognosis |
| Undifferentiated tumor /Anaplasia | Lack of differentiation: No any resemblance to normal tissue | Requires histology, IHC or molecular tests for classification<br>Often correlates with aggressive behavior |

# Microscopic features of malignancy:

- Marked pleomorphism: variation in cell and nuclear size/shape
- Hyperchromasia, darkly stained and enlarged nuclei
- Increased nuclear-to-cytoplasmic ratio (may approach 1 : 1 instead of the normal 1 : 4 or 1 : 6.)
- Tumor giant cells may be present
- Numerous and atypical mitoses: tripolar, quadripolar or multipolar spindles
- Loss of polarity and disorganized architecture



Well-differentiated squamous cell carcinoma of the skin. The tumor cells are strikingly similar to normal squamous epithelial cells, with intercellular bridges and nests of keratin pearls



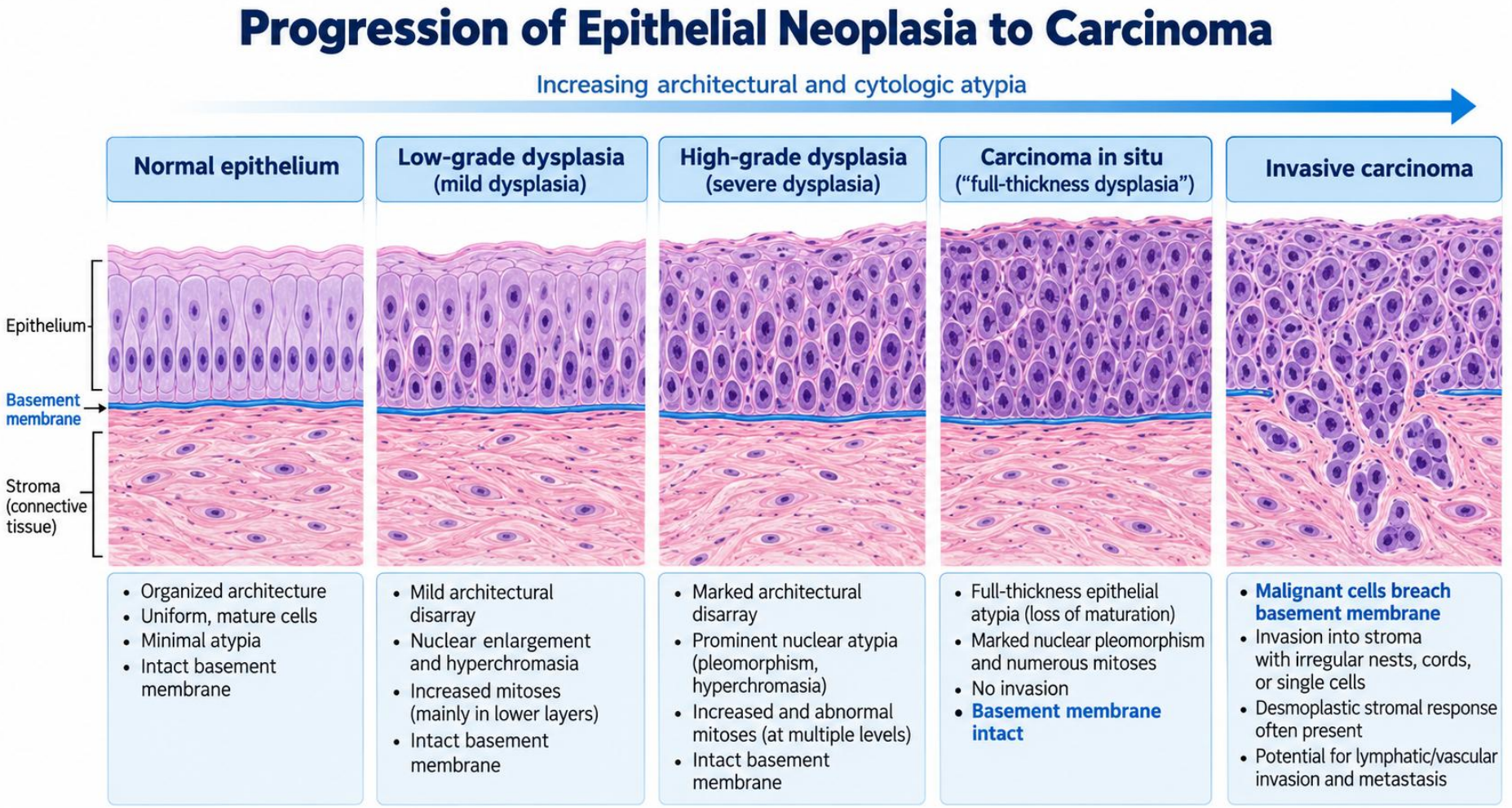
High-power detail view of malignant (anaplastic) tumor cells shows cellular and nuclear variation in size and shape with numerous atypical mitotic figures

# Terms used:

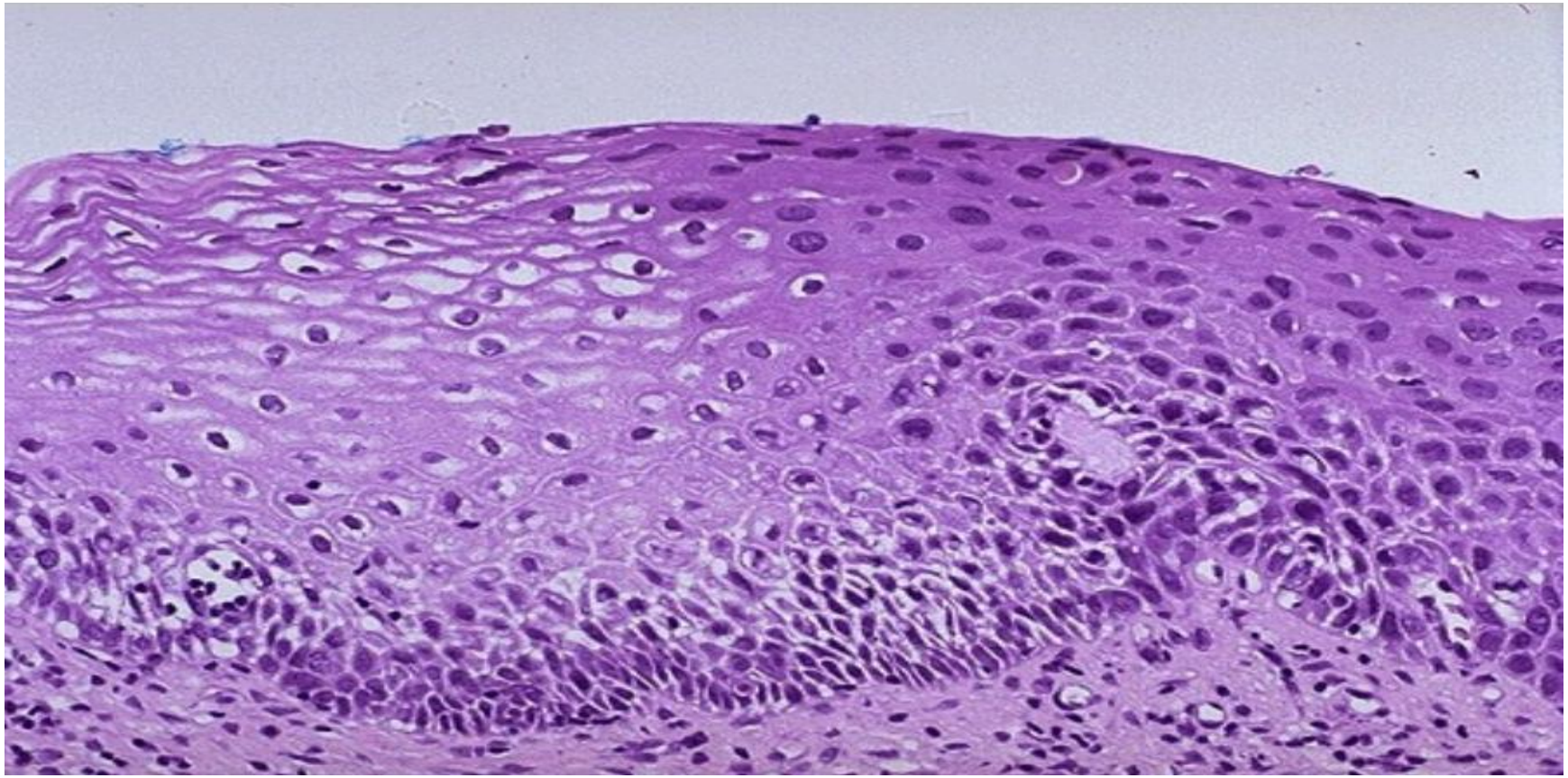
**Dysplasia:** The presence of abnormal (ATYPICAL) cells within a tissue or organ.
A term used to describe disorderly but non-neoplastic proliferation; it is encountered principally in the epithelia.
•Dysplasia is usually divided into low-grade dysplasia and high-grade dysplasia.

**Carcinoma in situ:** When dysplastic changes are marked and involve the **entire thickness of the epithelium**, the lesion is referred to as carcinoma in situ, a **pre-invasive** stage of cancer.

# Dysplasia → carcinoma in situ → invasion

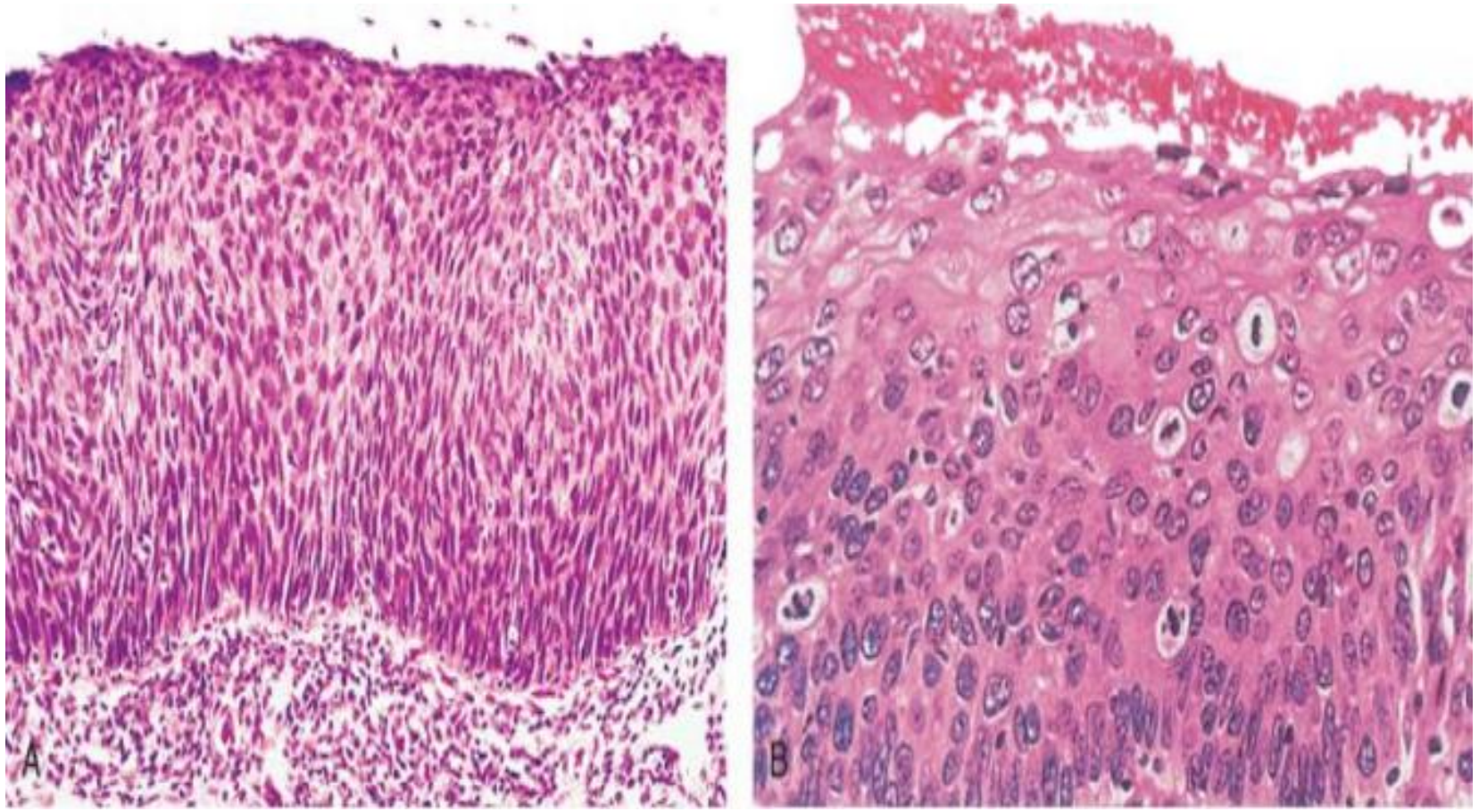


# Dysplasia



Dysplastic cells in the right compared to normal in the left

# Carcinoma in situ:



# Key distinction: in situ vs invasive carcinoma

| Lesion | Basement membrane | Why it matters |
|---|---|---|
| Dysplasia | Intact | Atypical cells with disordered maturation; may regress or progress |
| Carcinoma in situ | Intact | Full-thickness epithelial atypia; pre-invasive stage |
| Invasive carcinoma | Breached | Tumor has invaded stroma and can access vessels/lymphatics |

## Characteristics of benign and malignant neoplasms:
## 2. Rate of growth:

- Most benign tumors grow slowly, and most cancers grow much faster.
- The rate of growth of malignant tumors correlates in general with their level of differentiation (poorly differentiated tumors tend to be rapidly growing)

## Characteristics of benign and malignant neoplasms:
## 3. Local Invasion:

- A benign neoplasm remains localized at its site of origin
- Many benign neoplasms develop an enclosing fibrous capsule that separates them from host tissue.
- Malignant tumors grow by progressive infiltration and invasion of the surrounding tissue.
- They do not develop well-defined capsules.
- Next to the development of metastases, local invasiveness is the most reliable feature that distinguishes malignant from benign tumor

# Rate of growth and local invasion

| Feature | Benign | Malignant |
|---|---|---|
| Growth rate | Usually slow; may be hormone-dependent | Often faster, especially if poorly differentiated |
| Margins | Well circumscribed; often encapsulated | Poorly demarcated and infiltrative |
| Effect on adjacent tissue | Compresses but usually does not invade | Invades and destroys surrounding tissue |
| Invasion | Localized growth | Local invasiveness distinguishes malignancy from benign tumors |



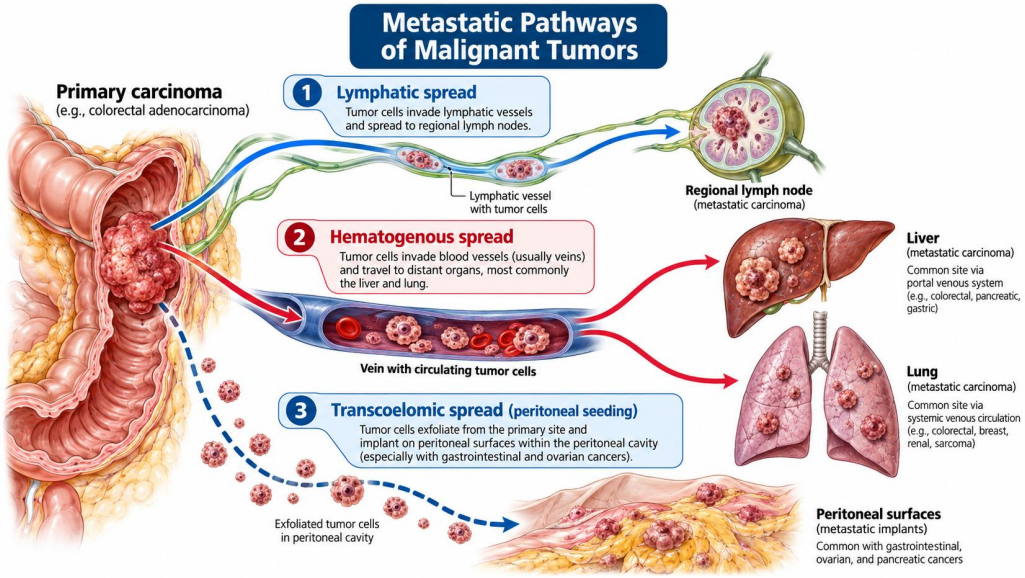
Fibroadenoma a benign tumor of the breast. The tan-colored, encapsulated small tumor is sharply demarcated from the whiter breast tissue

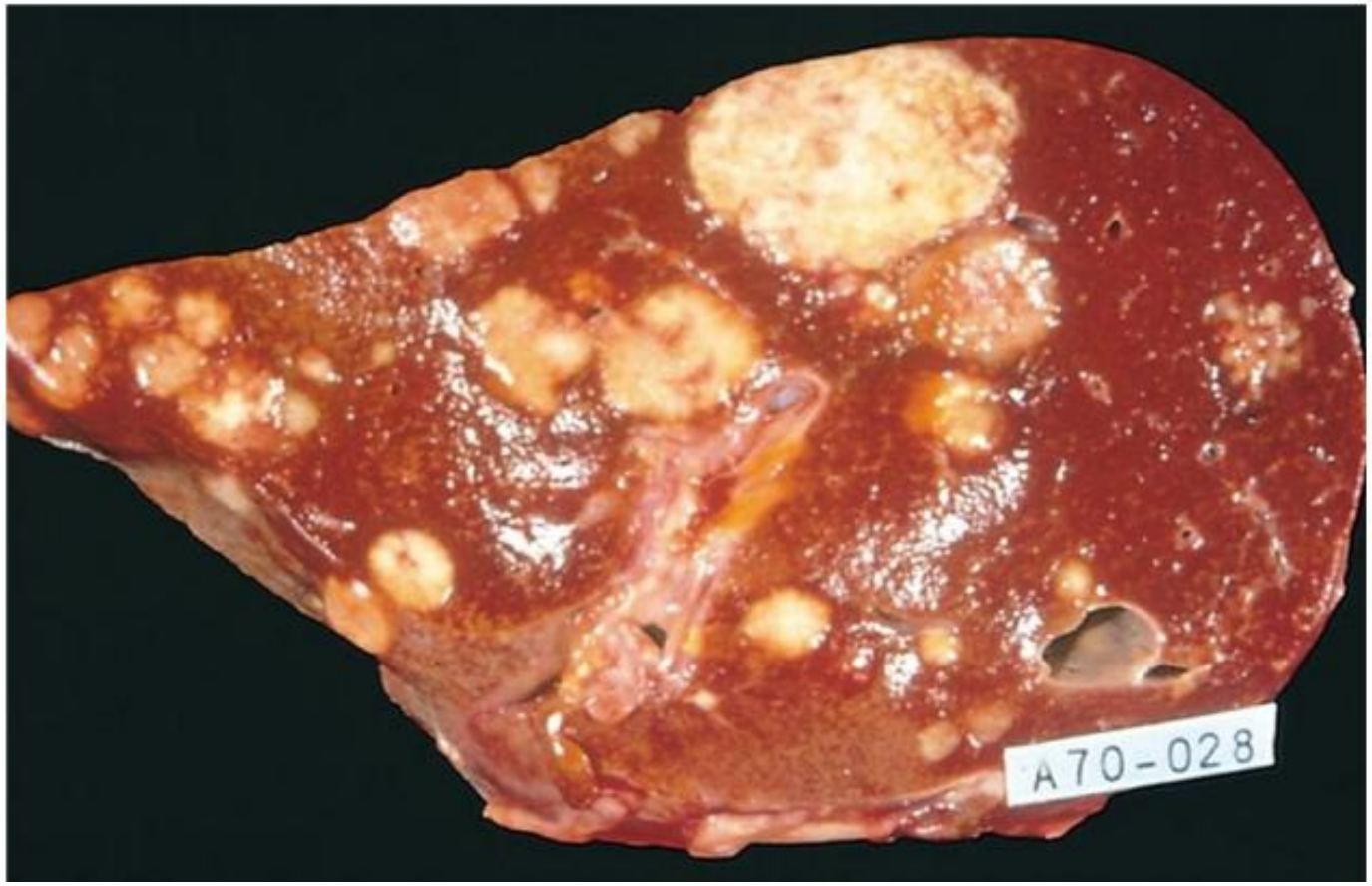



Cut section of invasive ductal carcinoma of the breast. The lesion is infiltrating the surrounding breast substance, and would be stony-hard on palpation

# Characteristics of benign and malignant neoplasms:
# 4. Metastasis: defining feature of malignancy

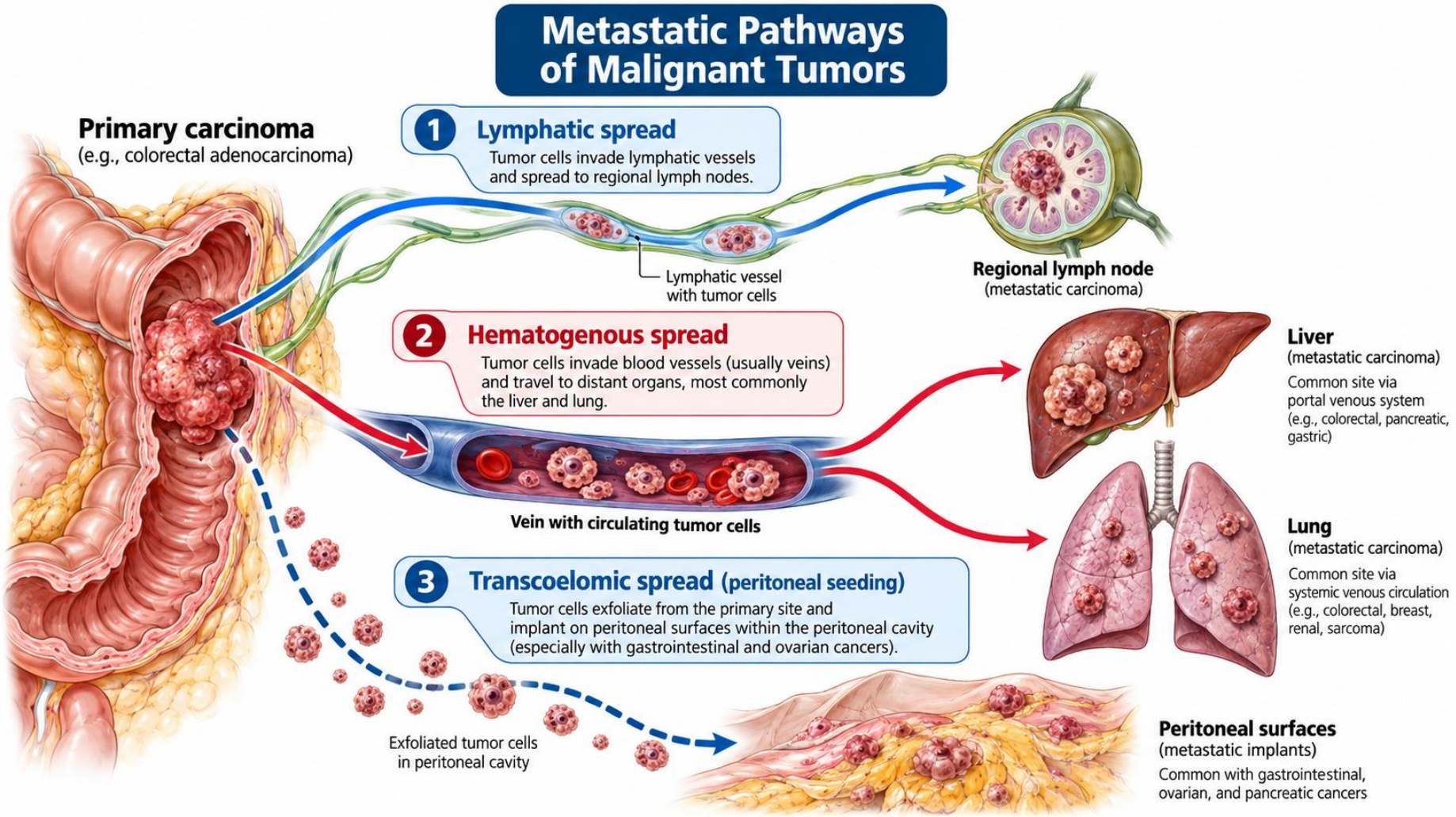
- Metastasis means secondary tumor implants discontinuous from the primary tumor.
- In general, larger and more anaplastic tumors are more likely to metastasize.
- Some malignant tumors rarely metastasize, such as basal cell carcinoma.






Liver with multiple metastatic nodules

# Pathways of malignant spread

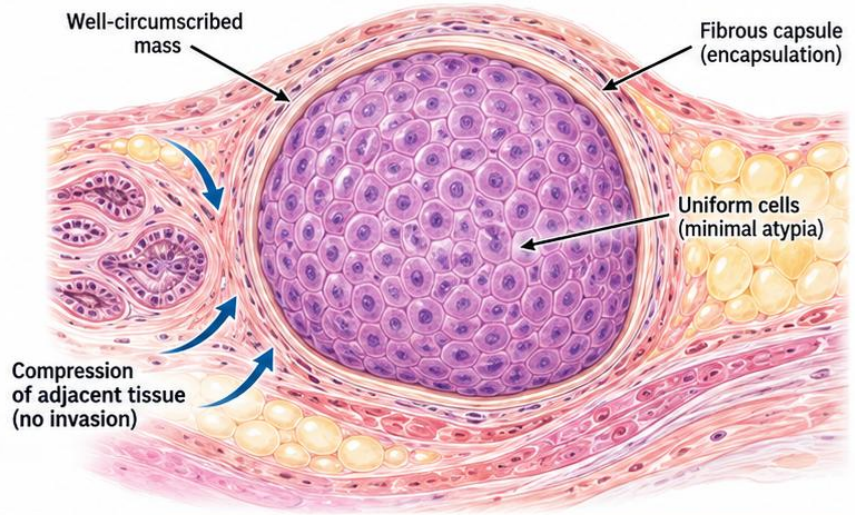


# Pathways of spread:

| Route | Main idea | Classic examples |
| --- | --- | --- |
| Lymphatic spread | Tumor cells enter lymphatics and spread to regional nodes | Common in carcinomas; sentinel lymph node biopsy assesses first draining node |
| Hematogenous spread | Tumor cells invade blood vessels, especially veins | Liver and lungs are common sites |
| Seeding body cavities | Tumor cells exfoliate into natural cavities | Ovarian carcinoma spreading over peritoneum |

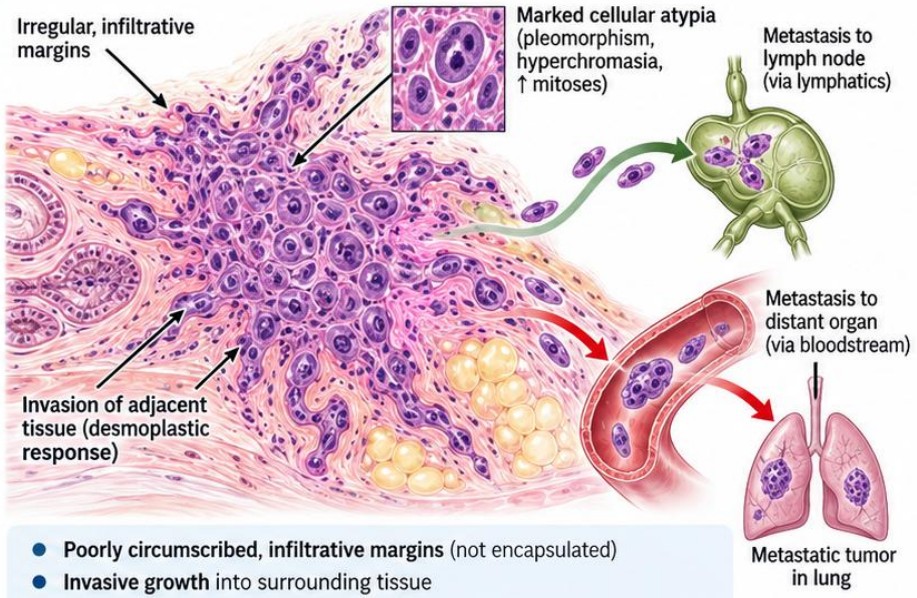
# Benign vs malignant neoplasms

## BENIGN NEOPLASM
Well-circumscribed, noninvasive, localized growth




- **Well circumscribed** and often encapsulated
- **Expansile growth** (compresses adjacent tissue)
- **Cytologically uniform cells** with **minimal atypia**
- **Slow growth**
- **Remains localized** (no metastasis)

## MALIGNANT NEOPLASM
Invasive growth, cellular atypia, metastasis

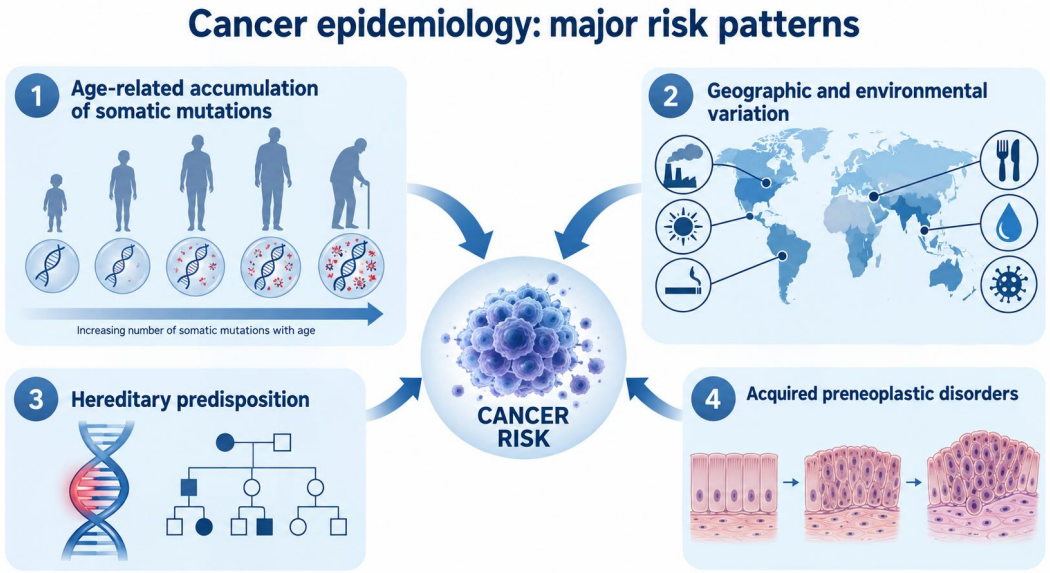



- **Poorly circumscribed, infiltrative margins** (not encapsulated)
- **Invasive growth** into surrounding tissue
- **Marked cellular atypia** (pleomorphism, hyper-chromasia, increased mitoses)
- **Can induce desmoplastic stromal response**
- **Potential for metastasis** (lymphatic and/or hematogenous spread)

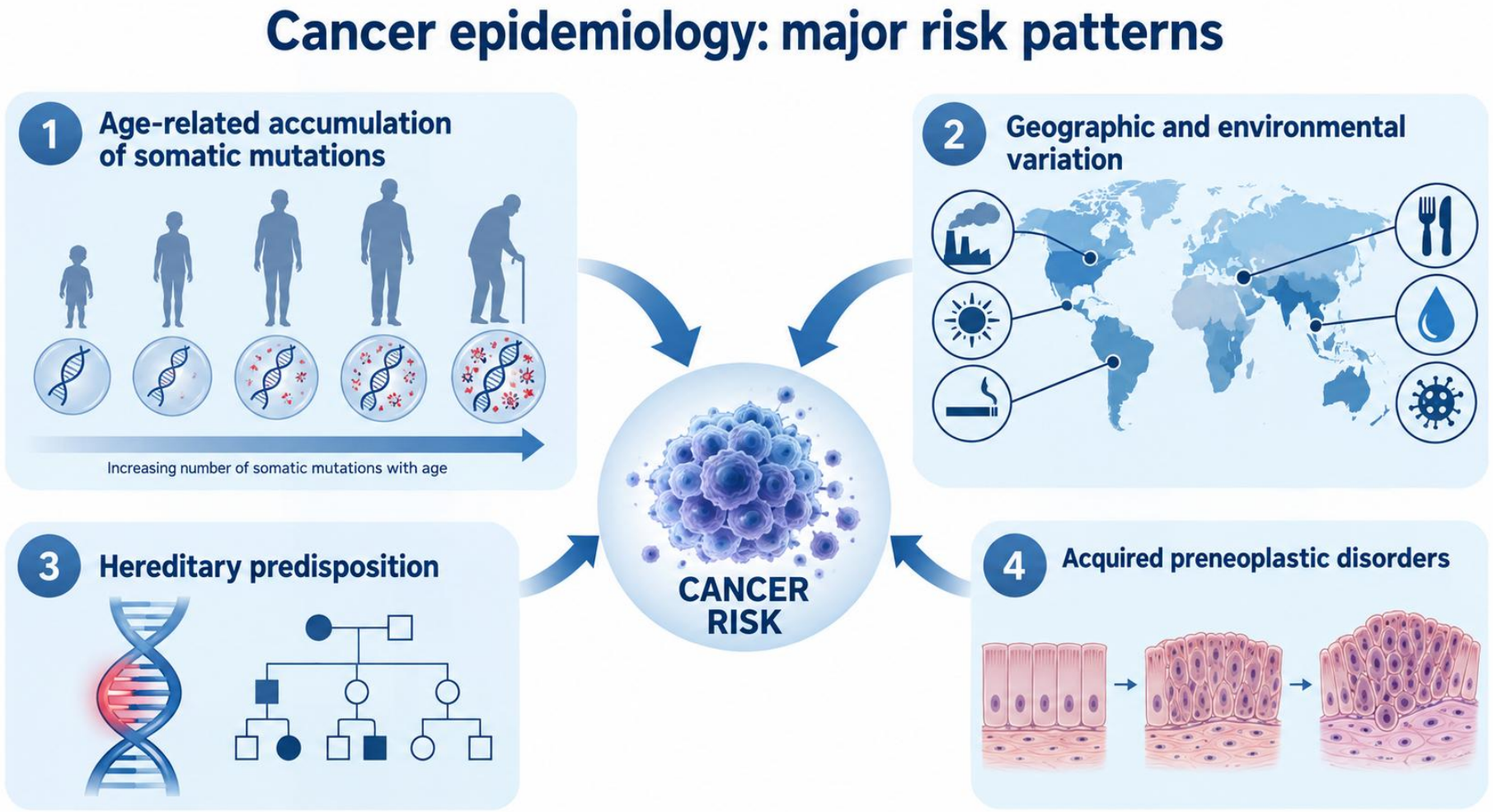
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# Cancer epidemiology: why it matters

- Epidemiologic patterns help identify environmental, hereditary and cultural risk factors.
- Cancer frequency varies with geography, age, inherited predisposition and acquired preneoplastic disorders.
- These patterns support prevention, screening and early diagnosis.



# Major epidemiologic risk patterns

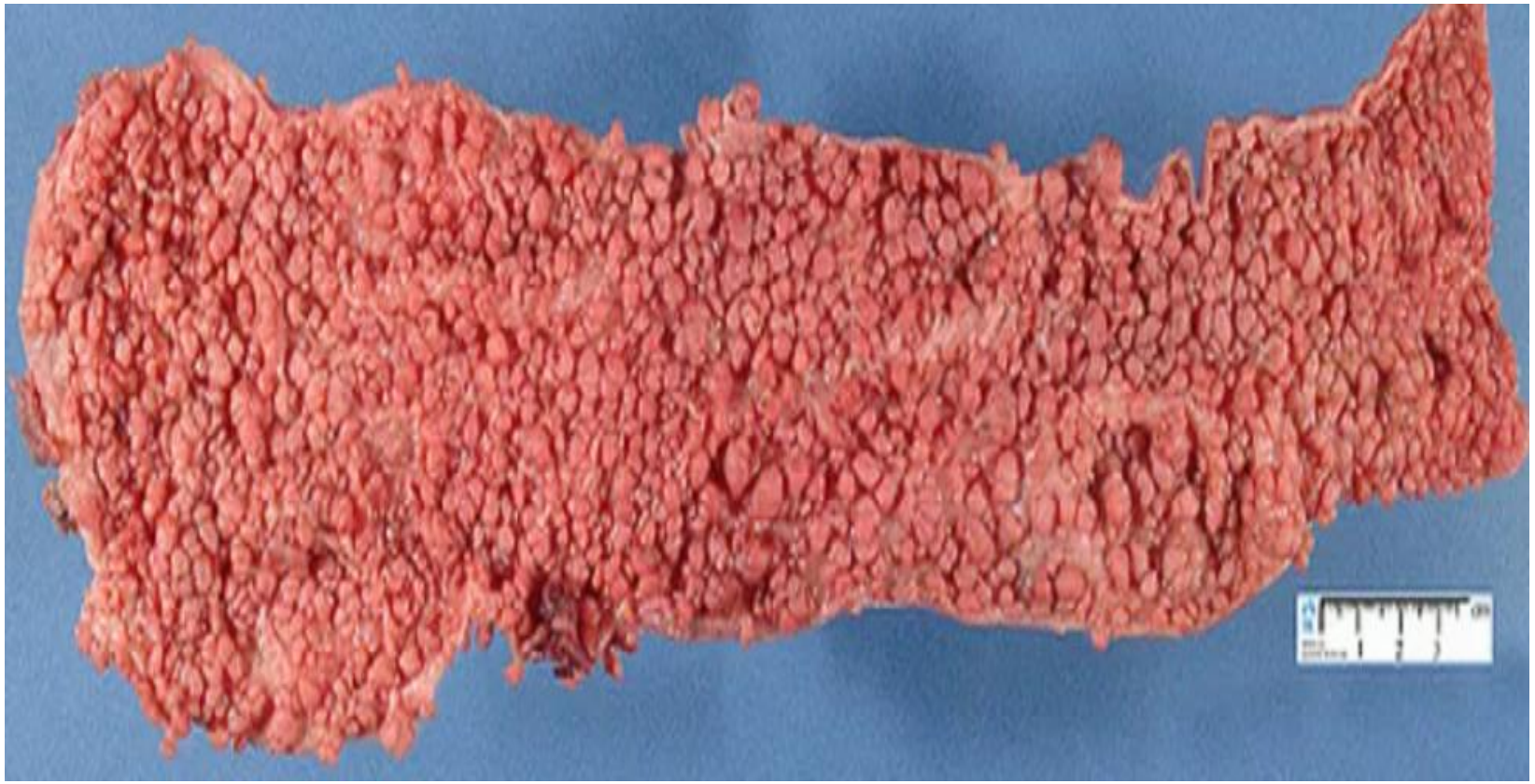


# Age and cancer type

| Age group | Common tumors / examples |
| --- | --- |
| Infancy and early childhood | Leukemia, lymphoma, neuroblastoma, nephroblastoma |
| Late childhood and adolescence | Osteosarcoma, Ewing tumor, rhabdomyosarcoma |
| Young adults | Testicular germ-cell tumors |
| Older adults | Colon and prostate cancers; overall cancer frequency rises with age |

# Heredity and inherited predisposition

| Category | Key point | Example |
| --- | --- | --- |
| Autosomal-dominant cancer syndromes | Often due to inherited tumor-suppressor gene mutation | Familial adenomatous polyposis: APC mutation |
| Defective DNA-repair syndromes | Failure to repair DNA damage increases mutation burden | HNPCC/Lynch syndrome: colon, endometrium, ovary |
| Familial cancers | Clustering without a clear single-gene pattern | Some breast and colon cancers in relatives |



Mutations in APC gene leads to development of hundreds or thousands of polyps in the colon and if not treated all patients develop colon cancer by the age of 50.

# Acquired preneoplastic disorders

| Disorder / condition | Associated malignancy |
|---|---|
| Long-standing skin fistula or chronic unhealed wound | Squamous cell carcinoma |
| Cirrhosis | Hepatocellular carcinoma |
| Atypical endometrial hyperplasia | Endometrial carcinoma |
| Dysplastic bronchial mucosa in smokers | Bronchogenic carcinoma |
| Chronic atrophic gastritis / H. pylori | Gastric carcinoma |
| Long-standing ulcerative colitis | Colorectal carcinoma |
| Oral leukoplakia | Squamous cell carcinoma |
| Villous adenoma of colon | Colorectal carcinoma |

# Summary

- Differentiation, growth rate, local invasion and metastasis are the core features used to distinguish benign from malignant tumors.
- Anaplasia indicates lack of differentiation and is a hallmark of malignancy.
- Carcinoma in situ is full-thickness epithelial atypia with an intact basement membrane; invasion begins when the basement membrane is breached.
- Metastasis occurs mainly by lymphatic spread, blood-borne spread or seeding of body cavities.
- Epidemiology links cancer risk to environment, age, heredity and preneoplastic disorders.

# Thank you

## Questions?